Get fields from the potentials

$$\vec{E}(\vec{r},t) = -\frac{\partial}{\partial t}\vec{A}(\vec{r},t) - \vec{\nabla}_r \varphi(\vec{r},t)$$

$$\vec{B}(\vec{r},t) = \vec{\nabla}_r \times \vec{A}(\vec{r},t)$$

Home work problem

$\Rightarrow$ need to learn how to differentiate functions of t' over t:

1) $$\frac{\partial R(t')}{\partial t} = \frac{\partial R(t')}{\partial t'}\frac{\partial t'}{\partial t} = -\frac{\vec{v}(t')\cdot\vec{R}(t')}{R(t')}\frac{\partial t'}{\partial t}$$

but $t' + \frac{R(t')}{c} = t$

$$\Rightarrow R(t') = c(t-t') \Rightarrow \frac{\partial R(t')}{\partial t} = c\left(1 - \frac{\partial t'}{\partial t}\right)$$

$$\Rightarrow c\left(1 - \frac{\partial t'}{\partial t}\right) = -\frac{\vec{v}\cdot\vec{R}}{R}\frac{\partial t'}{\partial t}$$

$$c = \left(c - \frac{\vec{v}\cdot\vec{R}}{R}\right)\frac{\partial t'}{\partial t}$$

$$\frac{\partial t'}{\partial t} = \frac{c}{c - \frac{\vec{v}\cdot\vec{R}}{R}} = \frac{1}{1 - \frac{\vec{v}\cdot\vec{R}}{cR}}$$

all variables on the right are functions of t'!

2) over $\vec{r}$: $$\vec{\nabla} R(t') = \vec{\nabla} c(t-t') = -c\,\vec{\nabla} t'$$

$$\vec{\nabla} t' = -\frac{1}{c}\vec{\nabla} R(t')$$

depends on $\vec{r}$ explicitly, $\vec{R} = \vec{r} - \vec{r}_0(t')$ and implicitly, via (t').

$$\vec{\nabla} R^2 = 2R\vec{\nabla}R$$

$$\parallel$$

$$\vec{\nabla}(\vec{r}-\vec{r}_0)^2 = \vec{\nabla}(\vec{r}^{\,2} - 2\vec{r}\cdot\vec{r}_0 + \vec{r}_0^{\,2}) = 2\vec{r} - 2\vec{r}_0$$

$$\Rightarrow \quad 2R\vec{\nabla}R = 2\vec{R} \Rightarrow \vec{\nabla}R = \frac{\vec{R}}{R}$$

Simply: $\vec{\nabla}R = \vec{\nabla}\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2} =$

$$= \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\sqrt{} =$$

$$= \hat{x}\,\frac{2(x-x_0)}{2\sqrt{}} + \hat{y}\,\frac{2(y-y_0)}{2\sqrt{}} + \hat{z}\,\frac{z-z_0}{\sqrt{}} = \frac{\vec{R}}{R}.$$

$$\Rightarrow \vec{\nabla}t' = -\frac{1}{c}\left\{\frac{\vec{R}}{R} + \underbrace{\frac{\partial R}{\partial t'}\vec{\nabla}t'}_{\equiv\,\vec{\nabla}R(t')}\right\}$$

$$\vec{\nabla}t' = -\frac{1}{c}\left\{\frac{\vec{R}}{R} - \frac{\vec{v}(t')\cdot\vec{R}(t')}{R(t')}\vec{\nabla}t'\right\}$$

$$\vec{\nabla}t'\left(1 - \frac{\vec{v}\cdot\vec{R}}{cR}\right) = -\frac{\vec{R}}{cR}$$

Finally $$\vec{\nabla}t' = -\frac{\vec{R}}{cR\left(1-\frac{\vec{v}\cdot\vec{R}}{cR}\right)} = -\frac{\vec{R}(t')}{c\left(R(t') - \vec{v}(t')\cdot\vec{R}(t')/c\right)}$$

$$\frac{\partial \vec{A}}{\partial t} = \left(\frac{q\mu_0}{4\pi}\right) \frac{\frac{\partial \vec{v}(t')}{\partial t'} \frac{\partial t'}{\partial t}}{R - \frac{\vec{v}\cdot\vec{R}}{c}} - \frac{\vec{v}(t')}{\left(R - \frac{\vec{v}\cdot\vec{R}}{c}\right)^2}\left[\frac{\partial R}{\partial t} - \frac{1}{c}\frac{\partial}{\partial t}(\vec{v}\cdot\vec{R})\right]$$

$$\frac{\partial R}{\partial t} = c\left(1 - \frac{\partial t'}{\partial t}\right) = c\left(1 - \frac{1}{1 - \frac{\vec{v}\cdot\vec{R}}{cR}}\right) = c\,\frac{-\frac{\vec{v}\cdot\vec{R}}{cR}}{1 - \frac{\vec{v}\cdot\vec{R}}{cR}} =$$

$$= -\frac{\vec{v}\cdot\vec{R}}{R - \frac{\vec{v}\cdot\vec{R}}{c}}$$

$$\frac{\partial \vec{R}}{\partial t} = \frac{\partial}{\partial t}\left(\vec{r} - \vec{r}_0(t')\right) = \frac{\partial}{\partial t'}\left(\vec{r} - \vec{r}_0(t')\right)\frac{\partial t'}{\partial t} =$$

$$= -\vec{v}(t')\frac{\partial t'}{\partial t} = -\frac{\vec{v}(t')}{1 - \frac{\vec{v}\cdot\vec{R}}{cR}} = -\frac{\vec{v}R}{R - \frac{\vec{v}\cdot\vec{R}}{c}}$$

$$\frac{\partial \vec{v}}{\partial t} = \frac{\partial \vec{v}}{\partial t'}\frac{\partial t'}{\partial t} = \frac{\dot{\vec{v}}}{1 - \frac{\vec{v}\cdot\vec{R}}{cR}} = \frac{\vec{a}R}{R - \frac{\vec{v}\cdot\vec{R}}{c}}$$

$$\frac{1}{c}\frac{\partial}{\partial t}(\vec{v}\cdot\vec{R}) = \frac{1}{c}\left(\frac{\dot{\vec{v}}\cdot\vec{R}}{1 - \frac{\vec{v}\cdot\vec{R}}{cR}} + \vec{v}\cdot\frac{\partial \vec{R}}{\partial t}\right) =$$

$$= \frac{\dot{\vec{v}}\cdot\vec{R} - \vec{v}^2}{c\left(1 - \frac{\vec{v}\cdot\vec{R}}{cR}\right)} = \frac{R}{c}\,\frac{\vec{a}\cdot\vec{R} - \vec{v}^2}{R - \frac{\vec{v}\cdot\vec{R}}{c}}$$

Thus $$\frac{\partial \vec{A}}{\partial t} = \left(\frac{\mu_0 q}{4\pi}\right)\left\{\frac{\dot{\vec{v}}}{R\left(1-\frac{\vec{v}\cdot\vec{R}}{cR}\right)^2} - \frac{\vec{v}}{R^2\left(1-\frac{\vec{v}\cdot\vec{R}}{cR}\right)^2}\right.$$ $\times$ usual multiplication

$$\left.\times\left[\frac{-\vec{v}\,\vec{R}}{R-\frac{\vec{v}\cdot\vec{R}}{c}} - \frac{\dot{\vec{v}}\cdot\vec{R}-\vec{v}^2}{c\left(1-\frac{\vec{v}\cdot\vec{R}}{cR}\right)}\right]\right\} = \left(\frac{\mu_0 q}{4\pi}\right)\cdot$$

$$\cdot\left\{\frac{\dot{v}R}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^2} + \frac{\vec{v}}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^2}\left[\frac{\vec{v}\cdot\vec{R}}{R-\frac{\vec{v}\cdot\vec{R}}{c}} + \frac{R}{c}\,\frac{\dot{\vec{v}}\cdot\vec{R}-\vec{v}^2}{R-\frac{\vec{v}\cdot\vec{R}}{c}}\right]\right.$$

$\mu_0 = \frac{1}{c^2\varepsilon_0}$ and let $\vec{a} \equiv \dot{\vec{v}} = \frac{\partial \vec{v}(t')}{\partial t'}$ is acceleration

$$\frac{\partial \vec{A}}{\partial t} = \frac{q}{4\pi\varepsilon_0 c^2}\,\frac{1}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left\{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)\vec{a}R + \right.$$

$$\left. + \vec{v}\left[\vec{v}\cdot\vec{R} + \frac{R}{c}\left(\vec{a}\cdot\vec{R}-\vec{v}^2\right)\right]\right\} =$$

$$= \frac{qc}{4\pi\varepsilon_0}\,\frac{1}{(cR-\vec{v}\cdot\vec{R})^3}\left\{(cR-\vec{v}\cdot\vec{R})\underbrace{\frac{R\vec{a}}{c}}_{\frac{R\vec{a}}{c}-\vec{v}+\vec{v}} + \vec{v}(\vec{v}\cdot\vec{R})\right.$$

$$\left. + \vec{v}\left(\vec{a}\cdot\vec{R}-v^2\right)\frac{R}{c}\right\} =$$

$$= \frac{qc}{4\pi\varepsilon_0}\,\frac{1}{(cR-\vec{v}\cdot\vec{R})^3}\left\{(cR-\vec{v}\cdot\vec{R})\left(-\vec{v}+\frac{R}{c}\vec{a}\right)\right.$$

$$\left. + \vec{v}\frac{R}{c}\left(c^2-v^2+\vec{a}\cdot\vec{R}\right)\right\}$$

Next $\vec{\nabla}\varphi(\vec{r},t) = \frac{q}{4\pi\varepsilon_0}\frac{-1}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^2}\vec{\nabla}\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)$

$$\vec{\nabla}R = -c\vec{\nabla}t' = -c\left(-\frac{\vec{R}}{c\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)}\right) = \frac{\vec{R}}{R-\frac{\vec{v}\cdot\vec{R}}{c}}$$

Use $\vec{\nabla}(\vec{v}\cdot\vec{R}) = \vec{v}\times(\vec{\nabla}\times\vec{R}) + \vec{R}\times(\vec{\nabla}\times\vec{v}) + (\vec{v}\cdot\vec{\nabla})\vec{R}$
$+ (\vec{R}\cdot\vec{\nabla})\vec{v}$.

1. $(\vec{R}\cdot\vec{\nabla})\vec{v} = \left(R_x\frac{\partial}{\partial x} + R_y\frac{\partial}{\partial y} + R_z\frac{\partial}{\partial z}\right)\vec{v}(t') =$

$$= R_x\frac{\partial\vec{v}}{\partial t'}\frac{\partial t'}{\partial x} + R_y\frac{\partial\vec{v}}{\partial t'}\frac{\partial t'}{\partial y} + R_z\frac{\partial\vec{v}}{\partial t'}\frac{\partial t'}{\partial z} =$$

$$= \vec{a}\left(\vec{R}\cdot\vec{\nabla}t'\right) = -\vec{a}\frac{\vec{R}^2}{c\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)}$$

2. $(\vec{v}\cdot\vec{\nabla})\vec{R} = (\vec{v}\cdot\vec{\nabla})(\vec{r}-\vec{r}_0(t')) = \vec{v} + \vec{v}\frac{\vec{v}\cdot\vec{R}/c}{R-\frac{\vec{v}\cdot\vec{R}}{c}} = \frac{\vec{v}R}{R-\frac{\vec{v}\cdot\vec{R}}{c}}$

since $(\vec{v}\cdot\vec{\nabla})\vec{r} = \vec{v}$

and $(\vec{v}\cdot\vec{\nabla})\vec{r}_0 = \left(v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)\vec{r}_0 =$

$$= v_x\frac{\partial\vec{r}_0}{\partial t'}\frac{\partial t'}{\partial x} + v_y\frac{\partial\vec{r}_0}{\partial t'}\frac{\partial t'}{\partial y} + v_z\frac{\partial\vec{r}_0}{\partial t'}\frac{\partial t'}{\partial z} =$$

$$= \vec{v}(t')\left(v_x\frac{\partial t'}{\partial x} + v_y\frac{\partial t'}{\partial y} + v_z\frac{\partial t'}{\partial z}\right) = \vec{v}(\vec{v}\cdot\vec{\nabla}t') =$$

$$= \vec{v}\left(\frac{-1}{c}\frac{\vec{v}\cdot\vec{R}}{R-\frac{\vec{v}\cdot\vec{R}}{c}}\right) = -\vec{v}\frac{\vec{v}\cdot\vec{R}/c}{R-\frac{\vec{v}\cdot\vec{R}}{c}}$$

3. $$\vec{\nabla}\times\vec{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\hat{x} + \ldots = \left(\frac{\partial v_z}{\partial t'}\frac{\partial t'}{\partial y} - \frac{\partial v_y}{\partial t'}\frac{\partial t'}{\partial z}\right)\hat{x} + \ldots$$

$$= -\vec{a}\times\vec{\nabla}t' = \frac{\vec{a}\times\vec{R}/c}{R - \frac{\vec{v}\cdot\vec{R}}{c}}$$

4. $$\vec{\nabla}\times\vec{R} = \vec{\nabla}\times(\vec{r}-\vec{r}_0) = -(-\vec{v}\times\vec{\nabla}t') = -\frac{\vec{v}\times\vec{R}/c}{R-\vec{v}\cdot\vec{R}/c}$$

$$\Rightarrow \vec{\nabla}(\vec{v}\cdot\vec{R}) = \frac{1}{R-\frac{\vec{v}\cdot\vec{R}}{c}}\left\{-\vec{v}\times\left(\vec{v}\times\frac{\vec{R}}{c}\right) + \vec{R}\times\left(\vec{a}\times\frac{\vec{R}}{c}\right)\right.$$

$$\left. - \vec{a}\,\frac{R^2}{c} \cancel{+\vec{v}\left(\vec{v}\cdot\frac{\vec{R}}{c}\right)} + \frac{\vec{v}R}{\cancel{R-\frac{\vec{v}\cdot\vec{R}}{c}}}\right\}$$

Use $A\times(B\times C) = B(A\cdot C) - C(A\cdot B)$

$$\Rightarrow \vec{v}\times(\vec{v}\times\vec{R}) = \vec{v}(\vec{v}\cdot\vec{R}) - \vec{R}v^2$$

$$\vec{R}\times(\vec{a}\times\vec{R}) = \vec{a}R^2 - \vec{R}(\vec{a}\cdot\vec{R})$$

so that $$\vec{\nabla}(\vec{v}\cdot\vec{R}) = \frac{1}{R-\frac{\vec{v}\cdot\vec{R}}{c}}\left\{v^2\frac{\vec{R}}{c} - \vec{v}\left(\vec{v}\cdot\frac{\vec{R}}{c}\right)\right.$$

$$\left. + \cancel{\vec{a}\frac{R^2}{c}} - \frac{\vec{R}}{c}(\vec{a}\cdot\vec{R}) - \cancel{\vec{a}\frac{R^2}{c}} \cancel{+\vec{v}\left(\vec{v}\cdot\frac{\vec{R}}{c}\right)}\right\}$$

$$+ \frac{\vec{v}R}{\cancel{R-\frac{\vec{v}\cdot\vec{R}}{c}}}$$

$$\Rightarrow \vec{\nabla}\varphi = \frac{q}{4\pi\varepsilon_0}\frac{-1}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left\{\vec{R} - \frac{1}{c}\vec{\nabla}(\vec{v}\cdot\vec{R})\right\} =$$

$$= \frac{q}{4\pi\varepsilon_0}\frac{+1}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left\{-\vec{R} + \frac{v^2}{c^2}\vec{R} - \frac{\vec{v}}{c}\left(\frac{\vec{v}}{c}\cdot\vec{R}\right)\right.$$

$$\left. - \vec{R}\,\frac{\vec{a}\cdot\vec{R}}{c^2} + \frac{\vec{v}R/c}{c}\right\}$$

$$\vec{\nabla}\varphi = \frac{q}{4\pi\varepsilon_0}\frac{1}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left\{\left(v^2-c^2-\vec{a}\cdot\vec{R}\right)\frac{\vec{R}}{c^2}\right.$$

$$\left.+\vec{v}\left(\frac{R}{c}-\frac{\vec{v}\cdot\vec{R}}{c^2}\right)\right\} =$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{qc}{\left(cR-\vec{v}\cdot\vec{R}\right)^3}\left\{\left(cR-\vec{v}\cdot\vec{R}\right)\vec{v}-\right.$$

$$\left.-\left(c^2-v^2+\vec{a}\cdot\vec{R}\right)\vec{R}\right\}$$

$$\Rightarrow \vec{E}=\frac{\partial\vec{A}}{\partial t}+\vec{\nabla}\varphi = \frac{qc}{4\pi\varepsilon_0\left(cR-\vec{v}\cdot\vec{R}\right)^3}\left\{\left(cR-\vec{v}\cdot\vec{R}\right)\left[\vec{v}+\right.\right.$$

$$\left.\left.-\vec{v}+\frac{R}{c}\vec{a}\right]+\left(c^2-v^2+\vec{a}\cdot\vec{R}\right)\left(\vec{v}\frac{R}{c}-\vec{R}\right)\right\} =$$

$$= \frac{qc}{4\pi\varepsilon_0}\frac{1}{\left(cR-\vec{v}\cdot\vec{R}\right)^3}\left\{\left(cR-\vec{v}\cdot\vec{R}\right)\frac{R\vec{a}}{c}\right.$$

$$\left.+\left(c^2-v^2+\vec{a}\cdot\vec{R}\right)\frac{\left(c\vec{R}-\vec{v}R\right)}{c}\right\} = \frac{1}{4\pi\varepsilon_0}\frac{q}{\left(cR-\vec{v}\cdot\vec{R}\right)^3}$$

$$\left\{-\left(c^2-v^2\right)c\left(\vec{R}-\frac{\vec{v}}{c}R\right)-\vec{a}\cdot\vec{R}\left(c\vec{R}-\vec{v}R\right)+\right.$$

$$\left.+R\vec{a}\left(cR-\vec{v}\cdot\vec{R}\right)\right\}$$

* the result

$$\vec{E} = \frac{q}{4\pi\varepsilon_0}\frac{1-v^2/c^2}{\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left(\vec{R}-\frac{\vec{v}}{c}R\right)$$

$$+\frac{q}{4\pi\varepsilon_0}\frac{1}{c^2\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)^3}\left[\vec{a}\cdot\vec{R}\left(\vec{R}-\frac{\vec{v}}{c}R\right)-\vec{a}R\left(R-\frac{\vec{v}\cdot\vec{R}}{c}\right)\right]$$

Consider $\vec{R} \times \left(\vec{R} - \frac{\vec{v}}{c} R\right) \times \vec{a} = \left(\vec{R} - \frac{\vec{v}}{c} R\right)\left(\vec{R} \cdot \vec{a}\right)$

$$- \vec{a}\left(\vec{R} \cdot \left(\vec{R} - \frac{\vec{v}}{c} R\right)\right) = \left(\vec{R} - \frac{\vec{v}}{c} R\right)\left(\vec{a} \cdot \vec{R}\right)$$

$$- \vec{a}\left(R^2 - R\,\frac{\vec{v} \cdot \vec{R}}{c}\right) =$$

$$= \left(\vec{R} - \frac{\vec{v}}{c} R\right) \vec{a} \cdot \vec{R} - \vec{a} R \left(R - \frac{\vec{v} \cdot \vec{R}}{c}\right)$$

Finally

$$\vec{E} = \frac{q}{4\pi\varepsilon_0} \left\{ \frac{1 - v^2/c^2}{\left(R - \frac{\vec{v} \cdot \vec{R}}{c}\right)^3} \left(\vec{R} - \frac{\vec{v}}{c} R\right) \right.$$

*

$$\left. + \frac{\vec{R} \times \left(\vec{R} - \frac{\vec{v}}{c} R\right) \times \vec{a}}{c^2 \left(R - \frac{\vec{v} \cdot \vec{R}}{c}\right)^3} \right\}$$

$\vec{a} \equiv \dot{\vec{v}}$ = acceleration.

Note: 1st term $\sim \frac{1}{R^2}$ and depends on $\vec{v}$.

2nd term $\sim \frac{1}{R}$ and ~~depend~~ is determined by acceleration $\vec{a}$.

Next $\vec{B}(\vec{r}, t) = \vec{\nabla}_r \times \vec{A}(\vec{r}, t)$

Calculate $\vec{B}$ via the integral form

$$\vec{A}(\vec{r},t) = \frac{\mu_0 q}{4\pi}\int dt' \, \frac{\vec{v}(t')}{R(t')} \, \delta\left(t' + \frac{R(t')}{c} - t\right)$$

$$R(t') = |\vec{r} - \vec{r}_0(t')| \quad , \quad \vec{R}(t') = \vec{r} - \vec{r}_0(t')$$

$$\vec{B}(\vec{r},t) = \vec{\nabla}_r \times \vec{A}(\vec{r},t) = \cancel{\nabla_R \vec{A} \times \nabla_r R}$$

$$= \sum_a \hat{e}_a \varepsilon^{abc} \frac{\partial}{\partial r_b} A_c = \sum_a \hat{e}_a \frac{\partial}{\partial R} \varepsilon^{abc} A_c \frac{\partial R}{\partial r_b} =$$

$$= \sum_a \hat{e}_a \left(-\varepsilon^{acb}\right) \frac{\partial A_c}{\partial R} \frac{\partial R}{\partial r_b} = -\frac{\partial \vec{A}}{\partial R} \times \vec{\nabla} R$$

$$\vec{\nabla} R = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$$

$$= \hat{x}\,\frac{2(x-x_0)}{2\sqrt{}} + \dots = \frac{\vec{R}}{R} = \hat{n} \text{ (unit vector)} = \hat{R}$$

$$\vec{B} = -\frac{\mu_0 q}{4\pi}\int dt' \left(\vec{v} \times \hat{R}\right) \frac{\partial}{\partial R} \frac{\delta\left(t' + \frac{R}{c} - t\right)}{R} =$$

$$= \frac{\mu_0}{4\pi}\int dt' \left(\vec{v} \times \hat{R}\right)\left(\frac{\delta\left(t' + \frac{R}{c} - t\right)}{R^2} - \frac{\delta'\left(t' + \frac{R}{c} - t\right)}{Rc}\right)$$

$$\delta' = \frac{\partial}{\partial t'}\delta()$$

But $\int dt' \, \delta\left(t' + \frac{R}{c} - t\right) f(t') =$

$$= \int d\tau \, \frac{\delta(\tau - t')}{1 - \frac{\vec{R}\cdot\vec{v}}{Rc}} f(\tau)$$

$$\cancel{\int dt \frac{\partial}{\partial t} \delta'\left(t' + \frac{R}{c} - t\right) = -\frac{\partial}{\partial t}\delta\left(t' + \frac{R}{c} - t\right)}$$

let $F(\tau) = \tau + \frac{R}{c} - t$ $\qquad dF = \frac{dF}{d\tau} d\tau$

$$\int d\tau \left(-\frac{\vec{v} \times \hat{R}}{Rc}\right) \frac{\partial \delta(F)}{\partial F} = \int dF \, \frac{1}{dF/d\tau} \, \frac{-\vec{v} \times \hat{R}}{Rc} \, \frac{\partial \delta(F)}{\partial F}$$

$$= \int dF\, \delta(F) \frac{d}{dF}\left(\frac{\vec{v}\times\hat{R}}{Rc}\,\frac{1}{dF/d\tau}\right) = \cancel{\int d\tau\, \frac{\delta(\tau-t')}{F'(\tau)}\, \vec{v}}$$

$$= \frac{1}{F'(\tau)}\,\frac{d}{d\tau}\left(\frac{\vec{v}\times\hat{R}}{Rc}\,\frac{1}{F'(\tau)}\right)\Bigg|_{F(\tau)=0} =$$

$$= \frac{1}{1-\frac{\vec{R}\cdot\vec{v}}{Rc}}\,\frac{\partial}{\partial t'}\,\frac{\vec{v}\times\hat{R}}{Rc-\vec{R}\cdot\vec{v}}$$

$$\Rightarrow \vec{B}(\vec{r},t) = \left(\frac{\mu_0 q}{4\pi}\right)\frac{\vec{v}\times\hat{R}}{R^2\left(1-\frac{\vec{v}\cdot\vec{R}}{cR}\right)} + \frac{Rc}{Rc-\vec{R}\cdot\vec{v}}\,\frac{\partial}{\partial t'}\left(\frac{\vec{v}\times\hat{R}}{Rc-\vec{R}\cdot\vec{v}}\right)$$

RHS is at t' such that $t' = t - \frac{R(t')}{c}$.

$$\frac{\partial\vec{v}(t')}{\partial t'} = \vec{a}(t'), \quad \frac{\partial R}{\partial t'} = -\frac{\vec{R}\cdot\vec{v}}{R}, \quad \frac{\partial\vec{R}}{\partial t'} = \frac{\partial}{\partial t'}\left(\vec{r}-\vec{r}_0(t')\right) = -\vec{v}$$

$$\frac{\partial\hat{R}}{\partial t'} = \frac{\partial}{\partial t'}\frac{\vec{R}}{R} = \frac{-\vec{v}}{R} - \frac{\vec{R}}{R^2}\left(-\frac{\vec{R}\cdot\vec{v}}{R}\right) = -\frac{\vec{v}}{R} + \frac{\vec{R}(\vec{R}\cdot\vec{v})}{R^3}$$

$$= \left(\vec{R}\times(\vec{R}\times\vec{v}) = \vec{R}(\vec{R}\cdot\vec{v}) - \vec{v}R^2\right) = \frac{\vec{R}\times(\vec{R}\times\vec{v})}{R^3}$$

$$\Rightarrow \vec{B}(\vec{r},t) = \left(\frac{\mu_0 q}{4\pi}\right)\frac{\vec{v}\times\vec{R}}{R^3\left(1-\frac{\vec{v}\cdot\vec{R}}{cR}\right)} +$$

$$+ \frac{Rc}{Rc-\vec{R}\cdot\vec{v}}\left\{\frac{1}{Rc-\vec{R}\cdot\vec{v}}\left(\vec{a}\times\frac{\vec{R}}{R} + \vec{v}\times\left[-\frac{\vec{v}}{R} + \frac{\vec{R}(\vec{R}\cdot\vec{v})}{R^3}\right]\right)\right.$$

$$\left. - \frac{\vec{v}\times\vec{R}}{R}\,\frac{-\frac{\vec{R}\cdot\vec{v}}{R}c - \vec{R}\cdot\vec{a} + \vec{v}^2}{(Rc-\vec{R}\cdot\vec{v})^2}\right\}$$

$$\cancel{\vec{B} = \left(\frac{\mu_0 q}{4\pi}\right) \frac{\left(1-\frac{v^2}{c^2}\right)(\vec{v}\times\vec{R})}{R^3}}$$

$$\vec{B} = \frac{\mu_0 q}{4\pi} \frac{\vec{v}\times\hat{R}}{R^2\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)} + \frac{1}{Rc\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^2}\left(\vec{a}\times\hat{R} + \frac{\vec{v}}{R}\times\hat{R}(\hat{R}\cdot\vec{v})\right) + \frac{(\vec{v}\times\hat{R})}{R^2c^2\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\left(c\,\vec{v}\cdot\hat{R} + \vec{a}\cdot\vec{R} - v^2\right)$$

$$= \frac{\mu_0 q}{4\pi} \frac{\vec{v}\times\hat{R}}{R^2\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)}\left\{1 + \frac{\vec{v}\cdot\hat{R}/c}{1-\frac{\vec{v}\cdot\hat{R}}{c}} + \frac{\vec{v}\cdot\hat{R}/c - v^2/c^2}{\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^2}\right\}$$

$$+ \left[\frac{\vec{a}\times\hat{R}}{Rc\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^2} + \frac{(\vec{v}\times\hat{R})\,\vec{a}\cdot\vec{R}}{R^2c^2\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\right]$$

but $\{\ \} = \dfrac{1}{1-\frac{\vec{v}\cdot\hat{R}}{c}} + \dfrac{\frac{\vec{v}\cdot\hat{R}}{c} - \frac{v^2}{c^2}}{\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^2} = \dfrac{1}{1-\frac{\vec{v}\cdot\hat{R}}{c}}\left\{1 + \dfrac{\frac{\vec{v}\cdot\hat{R}}{c} - \frac{v^2}{c^2}}{1-\frac{\vec{v}\cdot\hat{R}}{c}}\right\}$

$$= \frac{1 - v^2/c^2}{\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^2} \quad !$$

$$[\ ] = \frac{1}{Rc\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^2}\left[\vec{a}\times\hat{R} + \frac{\vec{a}\cdot\hat{R}\,(\vec{v}\times\hat{R})}{c\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)}\right] =$$

$$= \frac{\left(c - \vec{v}\cdot\hat{R}\right)\vec{a}\times\hat{R} + (\vec{a}\cdot\hat{R})(\vec{v}\times\hat{R})}{c^2R\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3} =$$

$$= \frac{\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)\vec{a} + (\vec{a}\cdot\hat{R})\frac{\vec{v}}{c}}{cR\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\times\hat{R}$$

$$\vec{B}(\vec{r},t) = \frac{-\mu_0 q}{4\pi}\,\hat{R}\times\left\{\frac{\left(1-v^2/c^2\right)\vec{v}}{R^2\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3} + \frac{\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)\vec{a}+(\vec{a}\cdot\hat{R})\frac{\vec{v}}{c}}{cR\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\right\}$$

but $-(\vec{v}\cdot\hat{R})\,\vec{a} + (\vec{a}\cdot\hat{R})\,\vec{v} = \hat{R}\times(\vec{a}\times\vec{v})$

$$= -\frac{\mu_0 q}{4\pi}\,\hat{R}\times\left\{\frac{\left(1-\frac{v^2}{c^2}\right)\vec{v}}{R^2} + \frac{\vec{a}+\hat{R}\times\left(\vec{a}\times\frac{\vec{v}}{c}\right)}{cR}\right\}\frac{1}{\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}$$

Take $\vec{E}$ from <u>p.13</u> now and evaluate

$$\hat{R}\times\vec{E} = \frac{q}{4\pi\varepsilon_0}\,\hat{R}\times\left\{\frac{\left(1-\frac{v^2}{c^2}\right)\left(\vec{R}-\frac{\vec{v}}{c}R\right)}{R^3\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3} + \frac{\vec{a}\cdot\vec{R}\left(\vec{R}-\frac{\vec{v}}{c}R\right)-\vec{a}R^2+\vec{a}R\frac{\vec{v}\cdot\vec{R}}{c}}{c^2R^3\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\right.$$

$$= \frac{q}{4\pi\varepsilon_0}\,\frac{1}{R\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\,\hat{R}\times\left\{\left(1-\frac{v^2}{c^2}\right)\frac{\left[\hat{R}-\frac{\vec{v}}{c}\right]}{R} + \frac{\vec{a}\cdot\hat{R}\left(\hat{R}-\frac{\vec{v}}{c}\right)}{c^2}\right.$$

$$\left. - \frac{\vec{a}}{c^2} + \frac{\vec{a}\,\vec{v}\cdot\hat{R}}{c^3}\right\} = \frac{q}{4\pi\varepsilon_0}\,\frac{1}{R\left(1-\frac{\vec{v}\cdot\hat{R}}{c}\right)^3}\cdot$$

$$\cdot\left\{\left(1-\frac{v^2}{c^2}\right)\frac{-\hat{R}\times\vec{v}}{cR} - \frac{\hat{R}\times\vec{a}}{c^2} - (\vec{a}\cdot\hat{R})\frac{\vec{R}\times\vec{v}}{c^3}\right.$$

$$\left. + \hat{R}\times\vec{a}\,\frac{\vec{v}\cdot\hat{R}}{c^3}\right\} \Rightarrow \text{all terms match!}$$

$$\cancel{\mu_0\varepsilon_0 A\frac{1}{c^2} = \mu_0\varepsilon_0 = \frac{A}{\varepsilon_0 c^2}}$$

$$\frac{1}{\varepsilon_0} = \frac{1}{\varepsilon_0\mu_0}\,\mu_0 = \mu_0 c^2 \quad \Rightarrow \boxed{\vec{B} = \frac{1}{c}\,\hat{R}\times\vec{E}}$$